FRAM

FM24W256 - 256Kbit (32K x 8) I2C FRAM Module

v1.0

2025-09-08
Rev. A

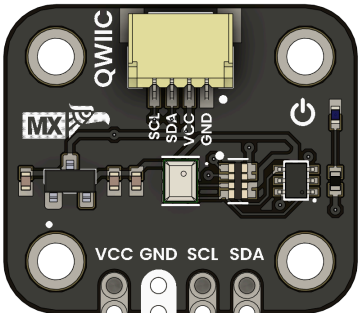
Professional electronic component

PRODUCT OVERVIEW

The FM24W256 module integrates a 256Kbit (32K x 8) I2C FRAM (Ferroelectric RAM) chip, providing a robust and efficient non-volatile memory solution for embedded applications. Designed for high-speed data access and low power operation, the module supports frequent and rapid read/write cycles with exceptional endurance. Unlike traditional EEPROM and other non-volatile memories, the FM24W256 ensures reliable data retention for up to 151 years, while minimizing system complexity and eliminating common reliability concerns. This makes it particularly suitable for data logging, configuration storage, and real-time data buffering in demanding embedded environments.

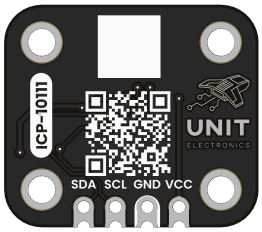
PRODUCT VIEWS

TOP VIEW



Component placement and connectors

BOTTOM VIEW



Underside components and connections

KEY TECHNICAL SPECIFICATIONS

🔌 CONNECTIVITY

Primary Interface:	**GPIO (Interrupt)**
Connector Type:	**JST 4-pin 1.0mm**
Logic Levels:	**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
`Power Supply`	3.3V or 5V
`Ground`	Common ground for all components

KEY FEATURES

Touch-only sensing
No physical press required – reacts to proximity of a finger.

Auto-calibration
Compensates for environmental changes and drift.

On-board pull-up/down
Ensures clean digital output.

JST PH-2.0 connector
Quick-disconnect cable interface.

Fast response
< 80 ms touch detection time.

Selectable modes
Momentary or toggle output (via solder-jumper on the board).

Mounting holes
Two M3 screw holes for easy panel integration.

ADDITIONAL TECHNICAL INFORMATION

📦 OVERVIEW

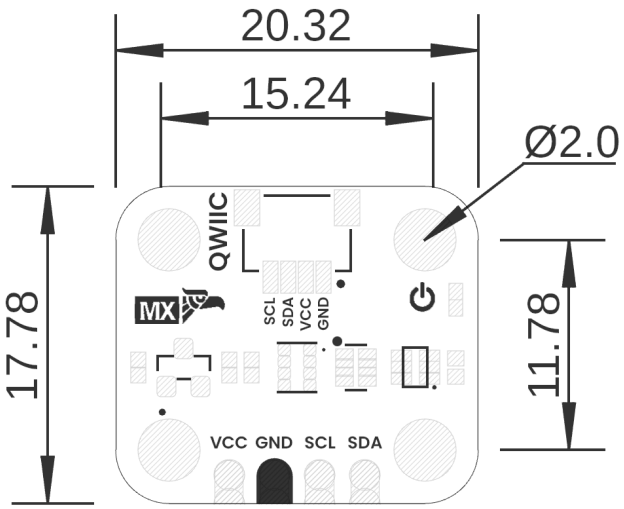
FEATURE	DESCRIPTION
Memory Type	FM24W256 I2C FRAM for dependable non-volatile storage
Interface	Standard I2C (2-wire) for straightforward integration
Operating Voltage	2.7V to 5.5V for flexible power compatibility
I2C Speed	Supports fast communication up to 400 kHz

TYPICAL APPLICATIONS

APPLICATION	DESCRIPTION
Data Logging	Ideal for storing sensor data and logs in real-time.
Configuration Storage	Retains system settings and configurations reliably.

HARDWARE DOCUMENTATION

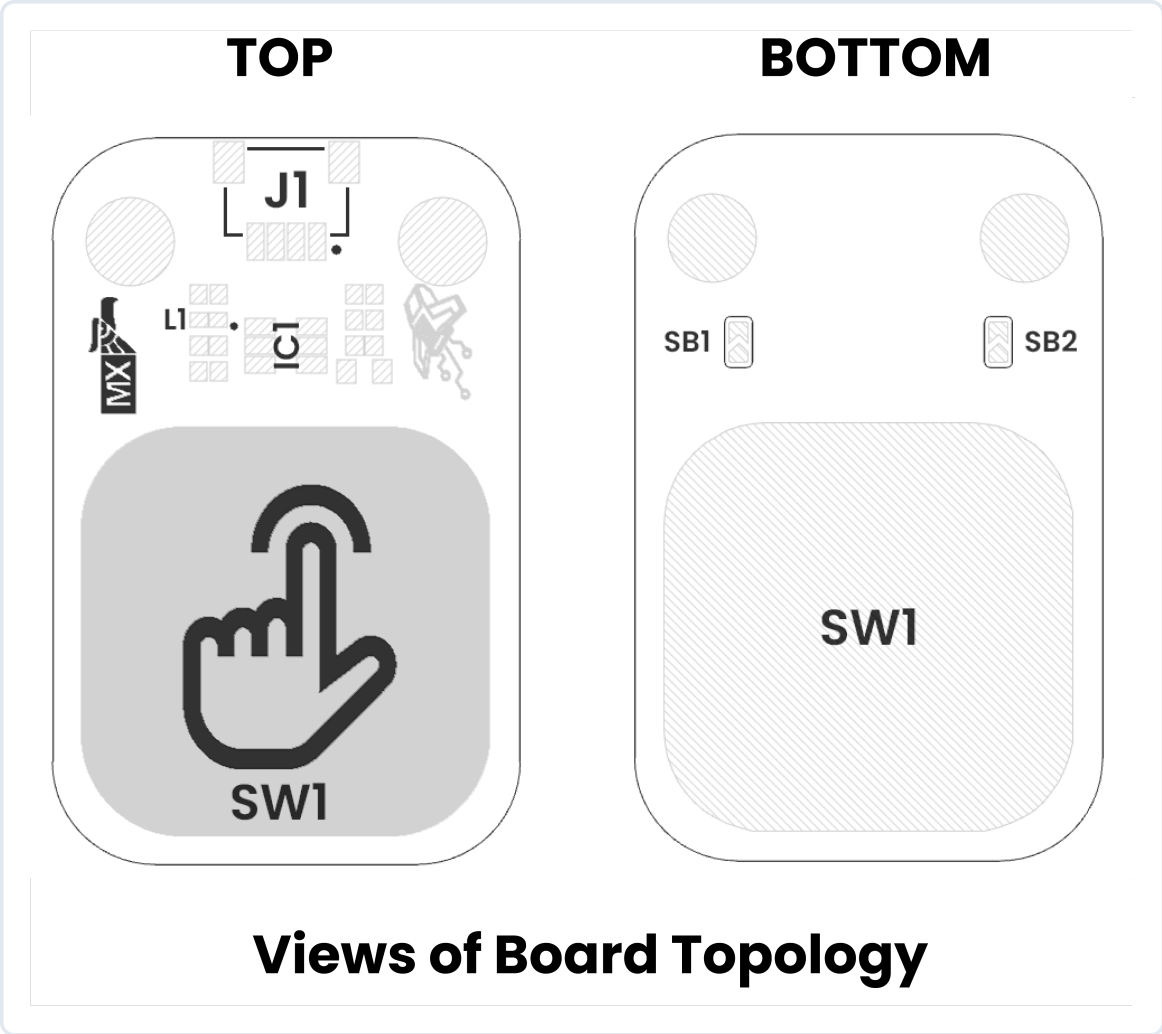
MECHANICAL DIMENSIONS



Mechanical dimensions in millimeters (mm)

Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



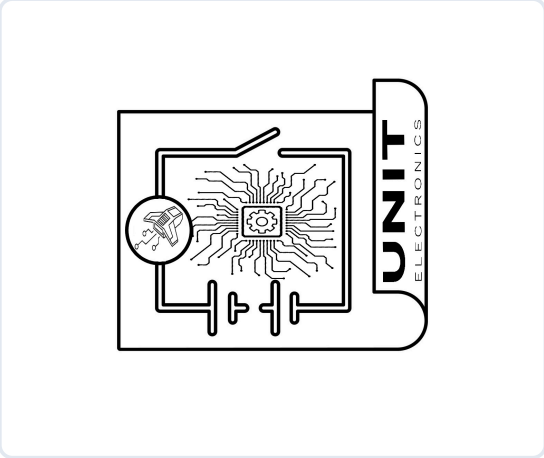
Connection topology and system integration diagram

Click image to open in full size

COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	{{sensor_description}}
L1	Power On LED
U1	{{regulator_description}}
JP1	2.54 mm Castellated Holes
J1	QWIIC Connector (JST 1 mm pitch) for I2C

CIRCUIT SCHEMATIC



Complete circuit schematic showing all component connections

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

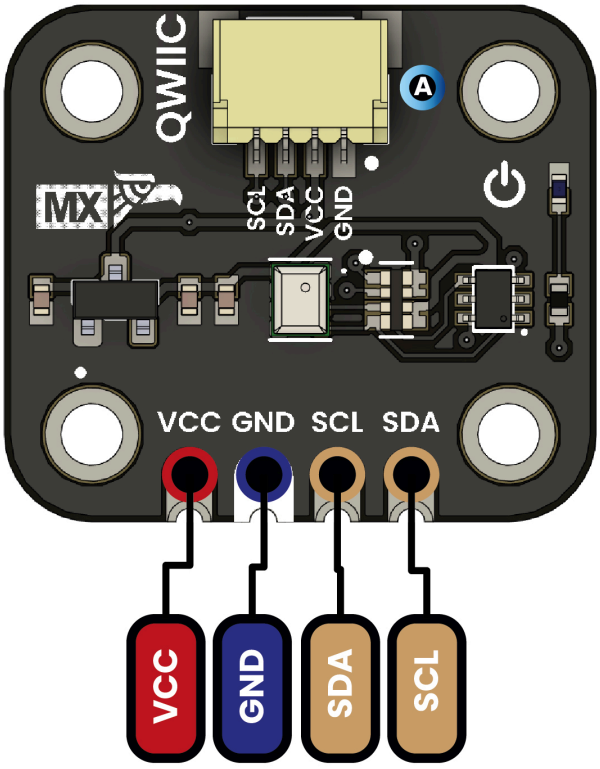
VOLTAGE LEVEL	FUNCTION
3.3 V – 5.5 V	Provides power to the on-board regulator and sensor core.
0 V	Common reference for power and signals.
1.8 V to VCC	Serial data line for I²C communications.
1.8 V to VCC	Serial clock line for I²C communications.

PIN CONFIGURATION LAYOUT

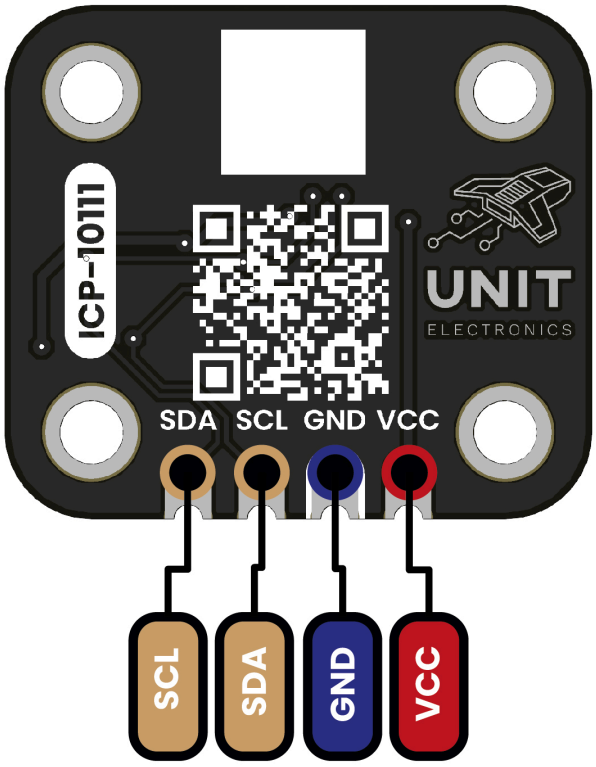
Physical connector layout and pin positioning

PINOUT

Top view



Bottom view



Description:

- Supply voltage
- GND
- I2C
- QWIIC

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

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